

Data Documentation

NFL Quarterback Retirement Research

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Research Overview

Purpose: Survival Analysis of Quarterback retirement in National Football League (NFL).

Built by: 00_nfl_EDA.Rmd

- **Event:** Retirement = last season in NFL (no games after season t).
- **Time:** Seasons since debut (**Career_Time**).
- **Censoring:** Active through end of data $\rightarrow$ Retired = 0. Data stops at 2016; ~98 QBs censored (EDA).

Raw inputs: Basic_Stats.csv, Game_Logs_Quarterback.csv, Career_Stats_Passing.csv

- Basic_Stats.csv:
- Game_Logs_Quarterback.csv:
- Career_Stats_Passing.csv:

Sources: [Kaggle NFL Statistics](#) (scraped from official [NFL](#) website)

Dataset:

File: 5_ALL_DATA.csv

Unit: Each row is one QB (Quarterback). **Scope:** QBs with 1 row in Game_Logs_Quarterback. **Merge:** Career + Debut-Year + First-Three-Year Summary Statistics

Note: No raw career totals used as predictors to avoid leaking survival time.

Load Data and Dimensions

```
nfl_data <- read_csv(
  here("nfl", "data", "cleaned", "5_ALL_DATA.csv"),
  show_col_types = FALSE)

nfl_data
```

```
# A tibble: 466 x 35
```

	Player_ID	Name	Retired	Debut_Age	`Height (inches)`	`Weight (lbs)`	First_Year
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	a.j.feel~	Feel~	1	25	75	216	2001
2	aaronrod~	Rodg~	0	22	74	225	2005
3	ajmccarr~	McCa~	0	24	75	220	2014
4	alanrish~	Rish~	1	25	74	190	1985
5	alexespi~	Espi~	1	24	73	193	1987
6	alexsmi~	Smit~	0	22	76	217	2005
7	alextann~	Tann~	0	26	76	220	2013
8	alexvanp~	Van ~	1	24	73	220	1993
9	alpastra~	Past~	1	26	73	190	1970
10	andrewar~	Ware~	1	22	74	205	1990

```
# i 456 more rows
```

```
# i 28 more variables: Career_Time <dbl>, Career_Games <dbl>,
#   Debut_Year_Games <dbl>, Three_Year_Games <dbl>, TD_INT <dbl>,
#   Debut_TD_INT <dbl>, Three_Year_TD_INT <dbl>, Career_Run_Pass <dbl>,
#   Debut_Run_Pass <dbl>, Three_Year_Run_Pass <dbl>,
#   Career_Completion_Percentage <dbl>, Debut_Completion_Percentage <dbl>,
#   Three_Year_Completion_Percentage <dbl>, Career_Yards_PG <dbl>, ...
```

```
# 35 columns, 466 rows (Quarterbacks)
dim(nfl_data)
```

```
[1] 466 35
```

Variable Reference

Identifiers and Outcome

Variable	Type	Description
Player_Id	character	Unique Player ID.
Name	character	Player name.
Retired	0/1	Outcome. 1 = last season < 2016; 0 = censored (active through 2016).

Demographics and Career Frame

Variable	Type	Description / formula
Debut_Age	numeric	Age at debut: $\text{Age} - (\text{2016} - \text{First_Year})$ (Original Data: Age = age in 2016).
Height (inches)	numeric	Height of Player in inches.
Weight (lbs)	numeric	Weight of Player in pounds.
First_Year	numeric	Debut year (~1970–2016).
Career_Time	numeric	Survival time: seasons from debut = $\text{Last_Year} - \text{First_Year} + 1$. Ex. 11 seasons - 2001 to 2011

Game counts (ONLY Regular-Season)

Variable	Description
Career_Games	Total Games Played in Career.
Debut_Year_Games	Total Games in debut season.
Three_Year_Games	Total Games in first 3 seasons.

Derived ratios (Career_* / Debut_* / Three_Year_*)

- **TD_INT:** TD Passes / max(Ints, 1) - Touchdown to Interception ratio; Infer accuracy/efficiency; higher → longer career (EDA).
- **RUN_PASS:** Rushing Yards / max(Passing Yards, 1) — mobility; can be <0 (sacks); higher → more hits, earlier retirement (EDA).

Completion and Per-Game stats (*_PG)

- **Completion_Percentage** (Career / Debut / Three-Year): Number of Throws Completions / Attempts; Infer accuracy/efficiency
- **Per-Game Statistics:** *_PG, Debut_*_PG, Three_Year_*_PG for Yards (passing), Sacks, TD (touchdowns), Ints (interceptions), Rush (rushing yards).
 - **Denominator** = Career_Games, Debut_Year_Games, or Three_Year_Games;
* **NA** when denominator = 0.

Missingness and Ranges

Missingness: Debut stats have more NAs (when there are no debut games or 0 attempts). Three-year window has fewer NAs.

- Most missingness is 'Debut_Completion_Percentage' with 181 of 466 rows NA.

```
missingness <- colSums(is.na(nfl_data))
missingness[missingness > 0]
```

Debut_Age	Career_Completion_Percentage
12	88
Debut_Completion_Percentage	Three_Year_Completion_Percentage
181	104
Career_Yards_PG	Debut_Yards_PG
78	155
Three_Year_Yards_PG	Career_Sacks_PG
90	78
Debut_Sacks_PG	Three_Year_Sacks_PG
155	90
Career_TD_PG	Debut_TD_PG
78	155
Three_Year_TD_PG	Career_Ints_PG
90	78
Debut_Ints_PG	Three_Year_Ints_PG
155	90
Career_Rush_PG	Debut_Rush_PG
78	155
Three_Year_Rush_PG	
90	

```
max(missingness)
```

```
[1] 181
```

Ranges:

- First_Year (Debut Year): 1970–2016
- RUN_PASS can be negative
- Many zeros in TD_INT / per-game for short careers. Consider filtering e.g. Big_T > 2 or Total_Games > 0 for modeling.

Data Structure and Summary

```
glimpse(nfl_data)
```

```
Rows: 466
Columns: 35
$ Player_ID      <chr> "a.j.feeley/2504566", "aaronrodgers/2~
$ Name           <chr> "Feeley, A.J.", "Rodgers, Aaron", "Mc~
$ Retired        <dbl> 1, 0, 0, 1, 1, 0, 0, 1, 1, 1, 0, 1, 0~
$ Debut_Age      <dbl> 25, 22, 24, 25, 24, 22, 26, 24, 26, 2~
$ `Height (inches)` <dbl> 75, 74, 75, 74, 73, 76, 76, 73, 73, 7~
$ `Weight (lbs)`  <dbl> 216, 225, 220, 190, 193, 217, 220, 22~
$ First_Year     <dbl> 2001, 2005, 2014, 1985, 1987, 2005, 2~
$ Career_Time    <dbl> 11, 12, 3, 3, 1, 12, 4, 11, 1, 4, 5, ~
$ Career_Games   <dbl> 28, 142, 8, 19, 1, 141, 1, 31, 4, 14,~
$ Debut_Year_Games <dbl> 1, 3, 0, 16, 1, 9, 0, 0, 4, 4, 16, 0,~
$ Three_Year_Games <dbl> 7, 7, 8, 19, 1, 32, 1, 1, 4, 9, 48, 1~
$ TD_INT         <dbl> 0.9032258, 4.1250000, 3.0000000, 1.00~
$ Debut_TD_INT   <dbl> 2.00000000, 0.00000000, 0.00000000, 0~
$ Three_Year_TD_INT <dbl> 1.3333333, 1.0000000, 3.0000000, 1.00~
$ Career_Run_Pass <dbl> 0.010610654, 0.069079751, 0.036299766~
$ Debut_Run_Pass <dbl> 0.000000000, 0.107692308, 0.000000000~
$ Three_Year_Run_Pass <dbl> 0.005199307, 0.142857143, 0.036299766~
$ Career_Completion_Percentage <dbl> 0.5577428, 0.6514924, 0.6638655, 0.59~
$ Debut_Completion_Percentage <dbl> 0.7142857, 0.5625000, NA, NA, 0.64285~
$ Three_Year_Completion_Percentage <dbl> 0.5714286, 0.5932203, 0.6638655, 0.59~
```

```

$ Career_Yards_PG          <dbl> 164.92857, 259.34507, 106.75000, 29.6~
$ Debut_Yards_PG           <dbl> 143.00000, 21.66667, NA, 0.00000, 69.~
$ Three_Year_Yards_PG      <dbl> 164.85714, 47.00000, 106.75000, 29.68~
$ Career_Sacks_PG          <dbl> 1.5714286, 2.4014085, 1.5000000, 0.63~
$ Debut_Sacks_PG           <dbl> 0.0000000, 1.0000000, NA, 0.0000000, ~
$ Three_Year_Sacks_PG      <dbl> 1.0000000, 1.2857143, 1.5000000, 0.63~
$ Career_TD_PG             <dbl> 1.0000000, 2.0915493, 0.7500000, 0.15~
$ Debut_TD_PG              <dbl> 2.0000000, 0.0000000, NA, 0.0000000, ~
$ Three_Year_TD_PG         <dbl> 1.1428571, 0.1428571, 0.7500000, 0.15~
$ Career_Ints_PG           <dbl> 1.1071429, 0.5070423, 0.2500000, 0.15~
$ Debut_Ints_PG            <dbl> 1.0000000, 0.3333333, NA, 0.0000000, ~
$ Three_Year_Ints_PG       <dbl> 0.8571429, 0.1428571, 0.2500000, 0.15~
$ Career_Rush_PG           <dbl> 1.7500000, 17.9154930, 3.8750000, 3.8~
$ Debut_Rush_PG            <dbl> 0.0000000, 2.3333333, NA, 0.6250000, ~
$ Three_Year_Rush_PG       <dbl> 0.8571429, 6.7142857, 3.8750000, 3.89~

```

```
summary(nfl_data)
```

Player_ID	Name	Retired	Debut_Age
Length:466	Length:466	Min. :0.0000	Min. :21.00
Class :character	Class :character	1st Qu.:1.0000	1st Qu.:23.00
Mode :character	Mode :character	Median :1.0000	Median :24.00
		Mean :0.8176	Mean :24.13
		3rd Qu.:1.0000	3rd Qu.:24.00
		Max. :1.0000	Max. :35.00
			NA's :12

Height (inches)	Weight (lbs)	First_Year	Career_Time
Min. :70.00	Min. :178.0	Min. :1970	Min. : 1.000
1st Qu.:74.00	1st Qu.:205.0	1st Qu.:1981	1st Qu.: 2.000
Median :75.00	Median :215.0	Median :1993	Median : 4.000
Mean :74.64	Mean :214.4	Mean :1993	Mean : 5.856
3rd Qu.:76.00	3rd Qu.:224.0	3rd Qu.:2006	3rd Qu.: 9.000
Max. :80.00	Max. :250.0	Max. :2016	Max. :22.000

Career_Games	Debut_Year_Games	Three_Year_Games	TD_INT
Min. : 0.00	Min. : 0.000	Min. : 0.00	Min. :0.0000
1st Qu.: 1.00	1st Qu.: 0.000	1st Qu.: 1.00	1st Qu.:0.0000
Median : 16.00	Median : 2.000	Median : 7.00	Median :0.5736
Mean : 40.62	Mean : 4.137	Mean :13.53	Mean :0.6791
3rd Qu.: 56.00	3rd Qu.: 7.000	3rd Qu.:25.00	3rd Qu.:1.0556
Max. :302.00	Max. :16.000	Max. :48.00	Max. :5.7500

Debut_TD_INT	Three_Year_TD_INT	Career_Run_Pass	Debut_Run_Pass
Min. :0.0000	Min. :0.000	Min. :-5.00000	Min. :-5.00000
1st Qu.:0.0000	1st Qu.:0.000	1st Qu.: 0.00000	1st Qu.: 0.00000
Median :0.0000	Median :0.375	Median : 0.02586	Median : 0.00000
Mean :0.3792	Mean :0.585	Mean : 0.06850	Mean : 0.18607
3rd Qu.:0.6000	3rd Qu.:1.000	3rd Qu.: 0.07355	3rd Qu.: 0.06526
Max. :6.0000	Max. :5.750	Max. : 6.85714	Max. :22.00000

Three_Year_Run_Pass	Career_Completion_Percentage	Debut_Completion_Percentage
Min. :-5.00000	Min. :0.0000	Min. :0.0000
1st Qu.: 0.00000	1st Qu.:0.4987	1st Qu.:0.4475
Median : 0.02125	Median :0.5480	Median :0.5093
Mean : 0.11521	Mean :0.5374	Mean :0.5075
3rd Qu.: 0.07845	3rd Qu.:0.5964	3rd Qu.:0.5769
Max. :17.00000	Max. :1.0000	Max. :1.0000
	NA's :88	NA's :181

Three_Year_Completion_Percentage	Career_Yards_PG	Debut_Yards_PG
Min. :0.0000	Min. : 0.00	Min. : -4.00
1st Qu.:0.4805	1st Qu.: 39.58	1st Qu.: 16.75
Median :0.5347	Median : 96.31	Median : 60.50
Mean :0.5276	Mean :104.27	Mean : 83.75
3rd Qu.:0.5865	3rd Qu.:158.79	3rd Qu.:146.22
Max. :1.0000	Max. :283.74	Max. :273.38
NA's :104	NA's :78	NA's :155

Three_Year_Yards_PG	Career_Sacks_PG	Debut_Sacks_PG	Three_Year_Sacks_PG
Min. : 0.00	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.: 28.34	1st Qu.:0.6329	1st Qu.:0.2111	1st Qu.:0.4286
Median : 77.79	Median :1.3809	Median :1.0000	Median :1.1640
Mean : 95.31	Mean :1.3516	Mean :1.2812	Mean :1.2879
3rd Qu.:153.81	3rd Qu.:1.9781	3rd Qu.:2.0000	3rd Qu.:2.0000
Max. :279.32	Max. :7.5000	Max. :7.5000	Max. :7.5000
NA's :90	NA's :78	NA's :155	NA's :90

Career_TD_PG	Debut_TD_PG	Three_Year_TD_PG	Career_Ints_PG
Min. :0.00000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.09524	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.3333
Median :0.47083	Median :0.2500	Median :0.3481	Median :0.6653
Mean :0.55455	Mean :0.4198	Mean :0.4969	Mean :0.6484
3rd Qu.:0.92483	3rd Qu.:0.7500	3rd Qu.:0.8542	3rd Qu.:0.9303
Max. :2.09155	Max. :3.0000	Max. :2.2791	Max. :4.0000
NA's :78	NA's :155	NA's :90	NA's :78

Debut_Ints_PG	Three_Year_Ints_PG	Career_Rush_PG	Debut_Rush_PG
Min. :0.0000	Min. :0.0000	Min. : -6.000	Min. : -6.000
1st Qu.:0.0000	1st Qu.:0.2000	1st Qu.: 0.994	1st Qu.: 0.000

```

Median :0.6000   Median :0.6440   Median : 3.723   Median : 2.750
Mean   :0.6241   Mean   :0.6429   Mean   : 5.918   Mean   : 6.214
3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.: 8.369   3rd Qu.: 9.354
Max.   :4.0000   Max.   :4.0000   Max.   :42.720   Max.   :54.333
NA's   :155      NA's   :90       NA's   :78       NA's   :155
Three_Year_Rush_PG
Min.    :-6.0000
1st Qu.: 0.4721
Median : 3.3333
Mean    : 6.1655
3rd Qu.: 8.5000
Max.    :47.1786
NA's    :90

```

Limitations

Topic	Note
Retirement	Inferred (no games after season t). Misclassification possible (injury, etc.).
Censoring	Cutoff 2016; QBs still active if Retired = 0. Consider sensitivity to cutoff.
Leakage	Remove career totals (there for reference); per-game and ratios (debut/3-year windows) used so predictors don't encode career length.

Data Pipeline

- **Script:** 00_nfl_EDA.Rmd containing full cleaning, formulas, and visuals.
- **Final Output:** nfl/data/cleaned/5_ALL_DATA.csv.
- **Intermediates Files:** 1_cleaned_nfl_data.csv, 2_reorder_nfl_data.csv, 3_cleaned_nfl_debut.csv, 4_cleaned_nfl_3year.csv.